

Auditing for Bias: Fairness in Classifier Models and Trade-offs

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1 ABSTRACT

The ProPublica's story on COMPAS data set and algorithm and the rebuttal of the claims made by NorthePointe Inc. [1] raises a question about models and fairness. This report analyzes the two different definitions of fairness using various simple classifier models and dedicated fairness based classifiers. We check for trade-offs between the definitions of fairness, and find that attaining fairness on both regards is not possible. Fair Classifiers tend to remove that bias by tuning the predictions to achieve fairness with respect to the sensitive attribute. We find in this project that ingraining the concept of fairness in a classifier model eats away at the performance of such machine learning models. Thus, we logically infer that Machine Learning models might not be suited for use as a true yardstick in extremely sensitive application domains like these, and can only be used as an auxiliary support system for making decisions on distributing surplus resources to humans.

2 INTRODUCTION

COMPAS is a popular scoring algorithm that is used by the justice systems in the US for pretrial and sentencing in criminal cases. It scores accused individuals based on the decile score where 1-4 are identified as low, 5-7 as medium, and 8-10 as high risk. COMPAS scores these in three different aspects, viz. Risk of Recidivism, Violence, and Failure to Appear. [3]

This scoring tool came into scrutiny around May of 2016 after a study conducted by ProPublica which indicated a bias in the result when the scores were compared for individuals from different race-groups [3]. They used the data for 18,610 people from the Broward County records to analyze this data (COMPAS data set) [2]. Broward County relies on the scores to detain or release the accused defendant. ProPublica's story on the COMPAS algorithm raises questions about models and fairness. An algorithm that dictates an individual's in/eligibility towards a resource/penalty must be fair to each and every individual regardless of all their protected attributes.

There are different definitions of fairness, and we cannot achieve fairness that satisfies all the different definitions as can be shown by the experiments performed in this project. The Propublica's report deems COMPAS as unfair in terms of Opportunity cost. It falsely labels African-Americans as high risk more often than Caucasians, although given that they do not re-offend.

The key questions we answer through this project are:

- Retaining vs. Removing the protected attributes attributes (eg. Race, gender, genetic identifiers, etc.) from the training process of the ML models (Anti-classification).
 - Can we remove bias by removing the protected attributes?
 - What are the effects of doing this in the performance of simple ML models?
- Does ProPublica's claim hold true? Does COMPAS exhibit fairness in a different way?
- How do fairer models compare with the simpler ML models in COMPAS like score prediction?
 - Which model performs better?
 - Which model is fairer, shows less bias?
- Is there a trade-off between the fairness and accuracy in ML models?

The approach and evaluation methodology used to answer these questions is discussed at length in the Methods section below.

Along with this, we discuss whether the fairness definition we seek in a model like this is the fairness in Opportunity Cost or if it is Demographic Classification Parity.

The results we found through this experiment were clearly indicative of the fact that if an ML model is trained on a data set with imbalances and biases in it, the prediction results will also be biased. Upon comparing the models when they use vs. when they do not use ‘race’, a protected feature in the data, it is found that there is no significant difference in both, performance and bias in the outcomes. The fair classifiers seem to shift the bias to obtain much fairer prediction results, but they suffer from inaccurate prediction. In our analysis, we replicated ProPublica’s analysis on the COMPAS data set and saw it to be true even though the results looked fair in terms of classification parity .

3 METHOD

To initiate the experiments, the data set was sourced from ProPublica’s analysis [2]. It has 52 different columns pertinent to an individual’s case history and crimes along with the columns that mark the chronology and time based information of the sentences served by the individuals. To begin the analysis, we start with pre-processing on the data. This includes the removal of null data from the data set. The data for races other than African-American, and Caucasian are also removed to fit the scope of the project, as it can be seen that other races have minimal weights as compared to the data for these two races in the data set. Figure1

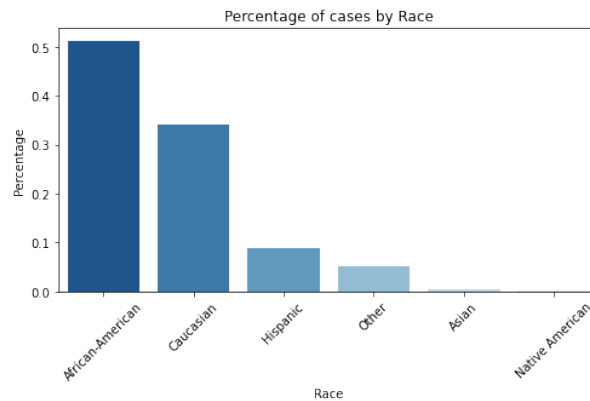


Fig. 1. Distribution by Race in the original Data.

3.1 Feature Selection

We select features that exhibit high correlation with the recidivism variable ‘two_year_recid’. When grouped by a feature, if the different groups are seen to differ by more than 10% of the two_year_recid value for one group, the feature is used in predicting recidivism. This change in ‘two_year_recid’ value indicates that the feature contains groups of values that correlate to different recidivism values. Such pivotal features were found to be: ‘race_factor’, ‘age_cat_encoded’, ‘sex_encoded’, ‘charge_degree_encoded’, ‘priors_count’.

3.2 Choice of and Analysis through Classifier Models

The data was divided as: 1/3rd of the data for the test set and the rest for the training set. For testing with a simple classifier, we used Decision Tree and Logistic Regression classifier models. We evaluated the Area under the curve (AUC) and Accuracies for both the models. Logistic Regression shows better promise with a higher Accuracy of above 0.64 Figure2. Logistic Regression was then used to classify the data and generate probabilities

False Positive Parity - Logistic Regression								High Risk	Low Risk
False Positive Parity Logistic Regression With Race Attribute					Data for Logistic Regression Without Race Attribute				
	Black	Probability	White	Probability		Black	Probability	White	Probability
Re-Offenders	198	0.34	39	0.74	Re-Offenders	337	0.0405	125	0.573
	336		412			230		168	
Non Re-Offenders	337	0.37	76	0.086	Non Re-Offenders	167	0.312	76	0.168
	194		217			367		375	

Table 1. Data and Probability Distribution for False Positive Parity with and without race using Logistic Regression

and distribution of prediction data 1 2. This was done firstly by keeping the protected race attribute in the data, and then by removing it to see the difference in fairness as seen on the left and right partitions on the table. The values pertaining to 'High Risk' have been highlighted in Red and 'Low Risk' in Blue.

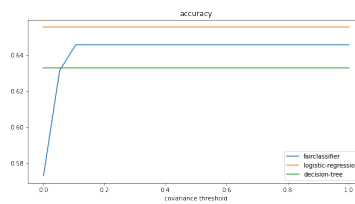


Fig. 2. Comparison of Accuracy from the Simple and Fair Classifier models.

3.3 Fairness Classifiers

Equal Opportunity Classifier and Demographic Parity Classifiers from sklearn by Scikit were used as the Fair Classifiers. In addition to the usual classification models, these fairness classifiers take in the bias sensitive attribute (i.e. protective attribute) as an input, and produce results that exhibit fairness with respect to the given attribute. Their fairness and accuracy performance were analyzed to see if they compare as better than Logistic Regression. Table 3 The values pertaining to 'High Risk' have been highlighted in Red and 'Low Risk' in Blue.

4 RESULTS

We observe by comparing the figures on the left and right side of Table 1 that there is no significant difference in the fairness of the simple models while retaining vs. while removing the protected attributes with respect to False Positive Parity. In fact, while removing the race attribute, the model that is otherwise 'fair' with respect to demographic parity becomes slightly biased. This is indicated by the slight disturbance in the data and probability distributions observed while comparing the left and right sides of Table 2. The rebuttal for COMPAS data set holds true as there is no significant change in probabilities for the two races in terms of Demographic Parity as seen on the right side of Table 2.

The Fair Classifier exhibits much fairer results when we compare it with the simpler model. As seen in Figure 3, there's roughly 25% increase in the p score and 30% increase in the equal opportunity score. From Table 3, we can see that there's a decrease in difference in false positives for the two groups while using Fair Classifiers when compared with simple classifiers Table 1. This indicates more fairness with fair classifiers.

In pursuit of bettering False positive Parity, the Demographic Calibration takes a hit, where probability difference is increased by 6.5% and 8% in Table 3 as opposed to the 1% and 2% difference in Table 2. This indicates that there is a trade-off between the two definitions. Along with this, from Table 2, it is also seen that although the fairness quotient is higher, the Fair models are less accurate as compared to simple models. This indicates the trade-off between fairness and accuracy as well.

Demographic Calibration - Logistic Regression								Will Re-Offend	Won't Re-Offend
Demographic Calibration Logistic Regression With Race Attribute					Data for Logistic Regression Without Race Attribute				
	Black	Probability	White	Probability		Black	Probability	White	Probability
High Risk	373	0.65	76	0.66	High Risk	337	0.66	125	0.62
	198		39			167		76	
Low Risk	194	0.63	217	0.65	Low Risk	230	0.61	168	0.69
	336		412			367		375	

Table 2. Data and Probability Distribution for Demographic Calibration Parity with and without race using Logistic Regression

False Positive Parity - Fair Classifier					Demographic Calibration - Fair Classifier				
	Black	Probability	White	Probability		Black	Probability	White	Probability
Re-Offenders	311	0.45	105	0.64	High Risk	311	0.68	105	0.6
	256		188			143		70	
Non Re-Offenders	143	0.26	70	0.15	Low Risk	256	0.604	188	0.669
	391		381			391		381	

Table 3. Data and Probability Distribution for different definitions of a Fair Classifier

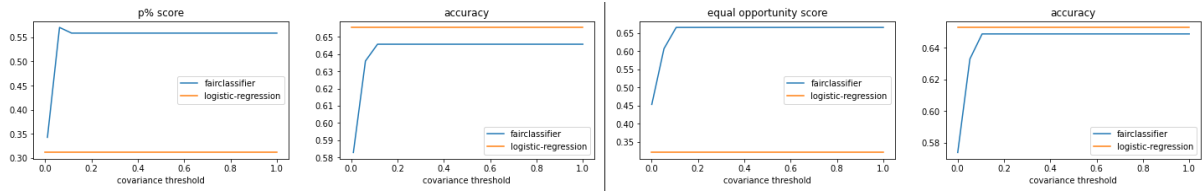


Fig. 3. p% score and Accuracy for Demographic Calibration | Equal Opportunity Score and Accuracy for False Positive Parity.

5 CONCLUSION

We were able to see from the findings in this project that merely retaining or removing the protected attributes do not impact the outcome of the learning models significantly, neither with respect to the bias, nor with respect to the performance of the model. ProPublica's statement against and the rebuttal for COMPAS algorithm, they are both true on their own. The algorithm widely used for pretrial evaluations is unfair with respect to the opportunity cost, but fair when their demographic calibration parity is concerned. The attainment of both the definitions of fairness is a mathematical impossibility. The fair models are indeed fair as shown by the p% and equal opportunity scores. They're also less but almost as accurate as the simpler models. It is observed that as the fairness increases, there is a decline in accuracy. This shows the trade-off between the two. The inherent bias introduced by the imbalance in the data for the two different races permeates onto the other attributes like violence, charge degree among others. It is important to reduce the False Positive rates entirely, even if it comes at the expense of accuracy which renders the model vain. Above all, in a domain like this where a False Positive has such a drastic impact on an individual's life, machine learning models shouldn't be accepted as the holy grail of fairness and justice. Possibly, in a different domain that deals with the distribution of surplus resources for preventative justice, models like these can be used as a reference to aid due human diligence.

REFERENCES

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